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Robotic surgical system, surgical robot, and robotic surgical method

Abstract

A robotic surgical system according to an embodiment may include: a surgical instrument including a pair of jaw members; an operation handle; and a surgical robot that includes a robot arm to which the surgical instrument is attached and which includes a drive part to drive a driven member, and a controller configured to drive the drive part based on a command jaw opening angle associated with an input to the input device for controlling an opening angle between the pair of jaw members. The controller is configured, when it is determined that a current value of the drive part exceeds a predetermined threshold value during a closing operation of the pair of jaw members, to drive the drive part in a restriction mode in which a magnitude of the command jaw opening angle is restricted.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Japanese Patent Applications No. 2021-138284 filed on Aug. 26, 2021 and No. 2022-128113 filed on Aug. 10, 2022, the entire contents of all of which are incorporated herein by reference.

BACKGROUND

(2) The disclosure may relate to a robotic surgical system, a surgical robot, and a robotic surgical method.

(3) In a related art, there has been known a robotic surgical system including a surgical instrument including a pair of jaw members.

(4) JP2020-536685 discloses a surgical robot system (a robotic surgical system) that includes a surgical tool including a pair of jaw members and a robot control system configured to control a grasping force and an angle between the pair of jaw members by using a position controller and a grasping force controller. For the grasping force controller to obtain (detect) tensions of cables that drive the pair of jaw members, the surgical robot system includes load cells or torque sensors. The grasping force controller controls, based on the tensions of the cables measured by the load cells or the torque sensors, the grasping force to a desired grasping force.

SUMMARY

(5) However, although the surgical robot system disclosed in JP2020-53665 can generate the desired grasping force, the grasping force controller needs to be provided with the load cells or the torque sensors to measure the tensions of the cables of the surgical tool. In such a case, there may be a problem that the structure around the cables (elongate elements) becomes complicated.

(6) An object of an embodiment of the disclosure may be to provide a robotic surgical system, a surgical robot, and a robotic surgical method that can generate an appropriate grasping force while suppressing a structure around elongate elements from being complicated.

(7) A first aspect of the disclosure may be a robotic surgical system that includes: a surgical instrument including a pair of jaw members configured to be driven by elongate elements connected to a driven member so as to be opened and closed to grasp an object; an input device to which a command for opening and closing the pair of jaw members is to be input; and a surgical robot that includes a robot arm to which the surgical instrument is attached and which includes a drive part configured to drive the driven member, and a controller configured to drive the drive part based on a command jaw opening angle associated with an input to the input device for controlling an opening angle between the pair of jaw members. The controller is configured to determine whether or not a current value of the drive part exceeds a predetermined threshold value during a closing operation of the pair of jaw members and, when it is determined that the current value of the drive part exceeds the predetermined threshold value, to drive the drive part in a restriction mode in which a magnitude of the command jaw opening angle is restricted.

(8) According to the first aspect, the controller is configured to determine whether or not the current value of the drive part exceeds the predetermined threshold value during the closing operation of the pair of jaw members, and when it is determined that the current value of the drive part exceeds the predetermined threshold value, to drive the drive part in the restriction mode in

which the magnitude of the command jaw opening angle is restricted. With this configuration, since the drive part is driven in the restriction mode in which the magnitude of the command jaw opening angle is restricted when it is determined that the motor current value of the drive part exceeds the predetermined threshold value during the operation of closing the pair of jaw members, it is possible to prevent the grasping force from becoming excessive. As a result, an appropriate grasping force can be obtained. Further, since the magnitude of the command jaw opening angle is restricted, it is not necessary to provide load cells or torque sensors to the elongate elements.

Therefore, it is possible to make a structure around the elongate elements less complicated than a case where the load cells or the torque sensors are provided to the elongate elements. As a result, it is possible to provide a robotic surgical system that is capable of providing an appropriate grasping force while preventing the structure around the elongate elements from being complicated.

(9) An second aspect of the disclosure may be a robotic surgical method for a robotic surgical system, wherein the robotic surgical system includes: a surgical instrument including a pair of jaw members configured to be driven by elongate elements connected to a driven member so as to be opened and closed to grasp an object; an input device to which a command for opening and closing the pair of jaw members is to be input; and a surgical robot including a robot arm to which the surgical instrument is attached and which includes a drive part configured to drive the driven member. The robotic surgical method may include: obtaining a command jaw opening angle that is associated with an input to the input device for controlling an opening angle between the pair of jaw members; and driving the drive part based on the command jaw opening angle. The driving of the drive part includes: determining whether or not a current value of the drive part exceeds a predetermined threshold value during a closing operation of the pair of jaw members; and when it is determined that the current value of the drive part exceeds the predetermined threshold value, driving the drive part in a restriction mode in which a magnitude of the command jaw opening angle is restricted.

(10) According to the second aspect, the driving of the drive part includes: determining whether or not the current value of the drive part exceeds the predetermined threshold value during the closing operation of the pair of jaw members; and when it is determined that the current value of the drive part exceeds the predetermined threshold value, driving the drive part in a restriction mode in which the magnitude of the command jaw opening angle is restricted. With this configuration, since the drive part is driven in the restriction mode in which the magnitude of the command jaw opening angle is restricted when it is determined that the motor current value of the drive part exceeds the predetermined threshold value during the operation of closing the pair of jaw members, it is possible to prevent the grasping force from becoming excessive. As a result, an appropriate grasping force can be obtained. Further, since the magnitude of the command jaw opening angle is restricted, it is not necessary to provide load cells or torque sensors to the elongate elements.

Therefore, it is possible to make a structure around the elongate elements less complicated than a case where the load cells or the torque sensors are provided to the elongate elements. As a result, it is possible to provide a robotic surgical method that is capable of providing an appropriate grasping force while preventing the structure around the elongate elements from being complicated.

(11) An third aspect of the disclosure may be a surgical robot that may include: a surgical instrument including a pair of jaw members configured to be driven by elongate elements connected to a driven member so as to be opened and closed to grasp an object; a robot arm to which the surgical instrument is attached and including a drive part configured to drive the driven member; a controller configured to drive the drive part based on a command jaw opening angle for an opening angle between the pair of jaw members. The controller is configured to determine whether or not a current value of the drive part exceeds a predetermined threshold value during a closing operation of the pair of jaw members, and when it is determined that the current value of the drive part exceeds the predetermined threshold value, to drive the drive part in a restriction mode in which a magnitude of the command jaw opening angle is restricted.

(12) According to the third aspect, the controller is configured to determine whether or not the current value of the drive part exceeds the predetermined threshold value during the closing operation of the pair of jaw members, and when it is determined that the current value of the drive part exceeds the predetermined threshold value, to drive the drive part in the restriction mode in which the magnitude of the command jaw opening angle is restricted. With this configuration, when it is determined that the motor current value of the drive part exceeds the predetermined threshold value during the closing operation of the pair of jaw members, the drive part is driven in the restriction mode in which the magnitude of the command jaw opening angle is restricted. Therefore, it is possible to prevent the grasping force from becoming excessive. As a result, an appropriate grasping force can be obtained. Further, since the magnitude of the command jaw opening angle is restricted, it is not necessary to provide load cells or torque sensors to the elongate elements. Therefore, it is possible to make a structure around the elongate elements less complicated than a case where the load cells or the torque sensors are provided to the elongate elements. As a result, it is possible to provide a surgical robot that is capable of providing an appropriate grasping force while preventing the structure around the elongate elements from being complicated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating a configuration of a robotic surgical system according to a first embodiment;
- (2) FIG. 2 is a block diagram illustrating a view of a control-related configuration of the robotic surgical system according to a first embodiment;
- (3) FIG. 3 is a diagram illustrating a view of a configuration of an operation handle according to a first embodiment;
- (4) FIG. 4 is a diagram illustrating a perspective view of a state where a surgical instrument is attached to a robot arm through an adaptor according to a first embodiment;
- (5) FIG. 5 is a diagram illustrating an exploded perspective view of a state where the surgical instrument is attached to a robot arm through an adaptor according to a first embodiment;
- (6) FIG. 6 is a diagram illustrating a perspective view of the surgical instrument according to a first embodiment as seen from below;
- (7) FIG. 7 is a diagram illustrating a perspective view of a state where a cover part is detached from a base part of the surgical instrument according to a first embodiment;
- (8) FIG. 8 is a diagram illustrating a plan view of a state where a cover part is detached from a base part of the surgical instrument according to a first embodiment;
- (9) FIGS. 9A and 9B are diagrams illustrating perspective views of an end effector of the surgical instrument according to a first embodiment;
- (10) FIG. 10 is an exploded perspective view illustrating a pair of jaw members and pulleys of the end effector according to a first embodiment;
- (11) FIG. 11 is a diagram illustrating the pair of jaw members, drive parts, and a controller according to a first embodiment;
- (12) FIG. 12 is a graph illustrating a relationship between a grip opening angle of the operation handle and a command jaw opening angle when there is no restriction according to a first embodiment;
- (13) FIG. 13 is a graph illustrating a relationship between the grip opening angle of the operation handle and the command jaw opening angle in the restriction mode according to a first embodiment;
- (14) FIG. 14 is a diagram illustrating an explanatory view for explaining grasping an object by the

pair of jaw members according to a first embodiment;

(15) FIG. 15 is a diagram of a cross-sectional view illustrating the elongate elements, pulleys, and a fall-off prevention part according to a first embodiment;

(16) FIG. 16 is a diagram of a flowchart illustrating a control process of starting the restriction mode in a state where the input device is operated according to a first embodiment;

(17) FIG. 17 is a diagram of a flowchart illustrating a control process of releasing (ending) the restriction mode in the state where the input device is operated according to a first embodiment;

(18) FIG. 18 is a diagram of a flowchart illustrating a control process of starting the restriction mode in a state where the input device is not operated according to a first embodiment;

(19) FIG. 19 is a diagram illustrating a pair of jaw members, drive parts, and a controller according to a second embodiment;

(20) FIG. 20 is a graph for explaining a restriction mode according to a second embodiment; and

(21) FIG. 21 is a graph illustrating a relationship between a detected angle and a minimum value of a command jaw opening angle according to a second embodiment.

DETAILED DESCRIPTION

(22) Descriptions are provided hereinbelow for one or more embodiments based on the drawings. In the respective drawings referenced herein, the same constituents are designated by the same reference numerals and duplicate explanation concerning the same constituents is omitted. All of the drawings are provided to illustrate the respective examples only.

First Embodiment

(23) (Configuration of Robotic Surgical System)

(24) A configuration of a robotic surgical system **100** according to a first embodiment is described with reference to FIGS. 1 and 2.

(25) As illustrated in FIG. 1, the robotic surgical system **100** includes a remote control apparatus **10** and a patient-side apparatus **20**.

(26) The remote control apparatus **10** is provided to remotely control medical equipment provided for the patient-side apparatus **20**. When an operator, as a surgeon, inputs an action mode instruction to be executed by the patient-side apparatus **20**, to the remote control apparatus **10**, the remote control apparatus **10** transmits the action mode instruction to the patient-side apparatus **20** through a controller **24**.

(27) In response to the action mode instruction transmitted from the remote control apparatus **10**, the patient-side apparatus **20** operates the medical equipment, including surgical instruments **40** attached to robot arms **21a** and an endoscope **50** attached to a robot arm **21b**. This allows minimally invasive surgery. Note that the patient-side apparatus **20** is an example of a surgical robot or a surgery assist robot. The controller **24** is an example of a controller or a control device.

(28) The patient-side apparatus **20** constitutes an interface to perform a surgery for a patient P. The patient-side apparatus **20** is positioned beside an operation table **30** on which the patient P is laid. The patient-side apparatus **20** includes plural robot arms **21a** and **21b**. One (**21b**) of the robot arms holds the endoscope **50** and the other robot arms (**21a**) hold the surgical instruments **40**. The robot arms **21a** and **21b** are commonly supported by an arm base **22**. Each of the plural robot arms **21a** and **21b** includes plural joints. Each joint includes a driver provided with a servo-motor and a position detector such as an encoder. The robot arms **21a** and **21b** are configured so that the medical equipment attached to each of the robot arms **21a** and **21b** is controlled by a driving signal given through the controller **24** and performs a desired movement.

(29) The arm base **22** is supported by a positioner **23** placed on the floor of an operation room. The positioner **23** includes a vertical articulated robot. The positioner **23** is configured to move the position of the arm base **22** three-dimensionally. The controller **24** is a control circuit including an arithmetic unit such as a CPU and/or the like, and a memory such as a ROM, a RAM, and/or the like.

(30) The surgical instruments **40** as the medical equipment are detachably attached to the distal

ends of the robot arms **21a**. Each surgical instrument **40** includes: a housing **41** (see FIG. **4**) which is to be attached to the robot arm **21a**; an elongated shaft **42** (see FIG. **4**); and an end effector **43** (see FIG. **4**) which is provided at a tip portion (a distal end portion) of the shaft **42**. The end effector **43** may be grasping forceps, scissors, a hook, a high-frequency knife, a snare wire, a clamp, a stapler, a clip applier, an electric knife, or a needle, for example. The end effector **43** is not limited to those and can be various types of treatment tools. In surgeries using the patient-side apparatus **20**, the robot arms **21a** introduce the surgical instruments **40** into the body of the patient P through a cannula (trocar) placed on the body surface of the patient P. The end effectors **43** of the surgical instruments **40** are then located near a surgery site.

(31) To the distal end of the robot arm **21b**, the endoscope **50** as the medical equipment is detachably attached. The endoscope **50** captures an image in a body cavity of the patient P. The captured image is outputted to the remote control apparatus **10**. The endoscope **50** may be a 3D endoscope capable of capturing a three-dimensional image or a 2D endoscope. In surgeries using the patient-side apparatus **20**, the robot arm **21b** introduces the endoscope **50** into the body of the patient P through a trocar placed on the body surface of the patient P. The endoscope **50** is then located near the surgery site.

(32) The remote control apparatus **10** constitutes an interface with the operator. The remote control apparatus **10** is an apparatus that allows the operator to operate the medical equipment attached to the robot arms **21a** and **22b**. Specifically, the remote control apparatus **10** is configured to transmit action mode instructions which are inputted by the operator and are to be executed by the surgical instruments **40** and endoscope **50**, to the patient-side apparatus **20** through the controller **24**. The remote control apparatus **10** is installed beside the operation table **30** so that the operator can see the condition of the patient P very well while operating the remote control apparatus **10**, for example. The remote control apparatus **10** may be configured to transmit action mode instructions wirelessly and installed in a room different from the operation room where the operation table **30** is installed.

(33) The action modes to be executed by the surgical instruments **40** include modes of actions to be taken by each surgical instrument **40** (a series of positions and postures) and actions to be executed by the function of each surgical instrument **40**. When the surgical instrument **40** is a pair of grasping forceps, for example, the action modes to be executed by the surgical instrument **40** include roll and pitch positions of the wrist of the end effector **43** and actions to open and close the jaws. When the surgical instrument **40** is a high-frequency knife, the action modes to be executed by the surgical instrument **40** include vibration of the high-frequency knife, specifically, supply of current to the high-frequency knife. When the surgical instrument **40** is a snare wire, the action modes to be executed by the surgical instrument **40** include a capturing action and an action to release the captured object. Further, the action modes may include an action to supply current to a bipolar or monopolar instrument to burn off the surgery site.

(34) The action modes to be executed by the endoscope **50** include the position and posture of the tip of the endoscope **50** and setting of the zoom magnification, for example.

(35) As illustrated in FIGS. **1** and **2**, the remote control apparatus **10** includes operation handles **11**, an operation pedal section **12**, a display **13**, and a control apparatus **14**. Note that the operation handle **11** is an example of an input device.

(36) The operation handles **11** are provided in order to remotely operate medical equipment attached to the robot arms **21a** and **21b**. Specifically, the operation handles **11** accept operations by the operator for operating the medical equipment (the surgical instruments **40** and endoscope **50**). The operation handles **11** are composed of two operation handles **11** arranged side by side in the horizontal direction. One of the two operation handles **11** is operated by the right hand of the operator while the other of the two operation handle **11** is operated by the left hand of the operator.

(37) The operation handles **11** extend from the rear side of the remote control apparatus **10** toward the front side. The operation handles **11** are configured to move in a predetermined three-

dimensional operation region. Specifically, the operation handles **11** are configured to be movable in a vertical direction, a horizontal direction, a front-rear direction, and a rotational direction.

(38) As illustrated in FIG. 3, the operation handles **11** are hand controllers to be operated by the hands of the operator. The operation handle **11** includes a support member **11a**, a pair of grip members **11b** provided on both sides of the support member **11a** with the support member **11a** being interposed between the pair of grip members **11b**, and a finger insertion portion **11c** provided in each of the pair of grip members **11b**. The operator inserts fingers (such as (thumb, middle finger, etc.) into the pair of finger insertion portions **11c** to operate the operation handle **11**. That is, a proximal end of each of the pair of grip members **11b** is rotatably connected to a support member **11a**. By increasing or decreasing an opening angle between the pair of grip members **11b** (a grip opening angle), the opening angle between a pair of jaw members **43a** and **43b**, which will be described later, is changed. A command for opening and closing the pair of jaw members **43a** and **43b** is input to the operation handle **11**. The opening angle between the pair of grip members **11b** is detected by a sensor, for example. For example, a hole sensor is provided in the support member **11a** of the operation handle **11** and a magnet is provided in each of one or both of the pair of grip members **11b** of the operation handle **11**, so that the opening angle between the pair of grip members **11b** can be detected. Or, a hole sensor is provided in one of the pair of grip members **11b** of the operation handle **11** and a magnet is provided in the other of the pair of grip members **11b** of the operation handle **11**, so that the opening angle between the pair of grip members **11b** can be detected. The signal regarding the detected opening angle between the pair of grip members **11b** is converted by a later-described controller **141** or the controller **24** to a command opening angle (a command jaw opening angle) regarding the opening angle θ between the pair of jaw members **43a** and **43b**.

(39) As illustrated in FIG. 1, the remote control apparatus **10** and patient-side apparatus **20** constitute a master-slave system in terms of controlling movements of the robot arms **21a** and the robot arm **21b**. The operation handles **11** constitute an operating part on the master side in the master-slave system, and the robot arms **21a** and **21b** holding the medical equipment constitute an operating part on the slave side. When the operator operates the operation handles **11**, the movement of one of the robot arms **21a** or **21b** is controlled so that the distal end portion (the end effector **43** of the surgical instrument **40**) of the robot arm **21a** or the distal end portion (the endoscope **50**) of the robot arm **21b** moves following the movement of the operation handles **11**.

(40) The patient-side apparatus **20** controls the movement of the robot arms **21a** in accordance with the set motion scaling ratio. When the motion scaling ratio is set to $\frac{1}{2}$, for example, the end effectors **43** of the surgical instruments **40** move $\frac{1}{2}$ of the movement distance of the operation handles **11**. This allows for precise fine surgery.

(41) The operation pedal section **12** or an operation pedal unit includes plural pedals to execute medical equipment-related functions. The plural pedals include a coagulation pedal, a cutting pedal, a camera pedal, and a clutch pedal. The plural pedals are operated by a foot of the operator.

(42) The coagulation pedal enables the surgical instrument **40** to coagulate a surgery site. Specifically, when the coagulation pedal is operated, voltage for coagulation is applied to the surgical instrument **40** to coagulate the surgery site. The cutting pedal enables the surgical instrument **40** to cut the surgery site. Specifically, the cutting pedal is operated to apply voltage for cutting to the surgical instrument **40** and cut a surgery site.

(43) The camera pedal is used to control the position and orientation of the endoscope **50** that captures images within the body cavity. Specifically, the camera pedal enables operation of the endoscope **50** by the operation handles **11**. That is, the position and orientation of the endoscope **50** are controllable by the operation handles **11** while the camera pedal is being pressed. The endoscope **50** is controlled by using both of the right and left operation handles **11**, for example. Specifically, when the operator rotates the right and left operation handles **11** about the middle point between the right and left operation handles **11**, the endoscope **50** is rotated. When the

operator presses the right and left operation handles **11** together, the endoscope **50** goes forward into the body cavity. When the operator pulls the right and left operation handles **11** together, the endoscope **50** goes back. When the operator moves the right and left operation handles **11** together up, down, right, or left, the endoscope **50** moves up, down, right, or left, respectively.

(44) The clutch pedal is used to temporarily disconnect operation-related connection between the operation handles **11** and the robot arms **21a** to stop movement of the surgical instruments **40**. Specifically, when the clutch pedal is being pressed, the robot arms **21a** of the patient-side apparatus **20** do not work even if the operation handles **11** are operated. For example, when the operation handles **11** are operated and moved to the edge of the range of movement, the operator operates the clutch pedal to temporarily disconnect the operation-related connection and then returns the operation handles **11** to the center of the range of movement. When the operator stops operating the clutch pedal, the operation handles **11** are again connected to the robot arms **21a**. The operator restarts the operation for the operation handles **11** around the center thereof.

(45) The display **13** (or a display device) is configured to display images captured by the endoscope **50**. The display **13** comprises a scope type display or a non-scope type display. (Note that FIG. **1** illustrates a scope type display **13**.) The scope type display is a display configured in such a manner that the operator looks into the display. The non-scope type display is a display like an open-type display that includes a flat screen and the operator is able to see without looking into, such as normal displays for personal computers.

(46) When the scope type display is attached, the scope type display displays 3D images captured by the endoscope **50** attached to the robot arm **21b** of the patient-side apparatus **20**. When the non-scope type display is attached, the non-scope type display also displays 3D images captured by the endoscope **50** provided for the patient-side apparatus **20**. The non-scope type display may display 2D images captured by the endoscope **50** provided for the patient-side apparatus **20**.

(47) The display **13** is provided with a head sensor **13a** for detecting whether or not the operation handle **11** is being operated by the operator. Specifically, the head sensor **13a** is configured to detect the presence of the head of the operator. The head sensor **13a** is configured to detect whether or not the operator looks into the display **13** to operate the operation handle **11**.

(48) As illustrated in FIG. **2**, the control apparatus **14** includes a controller **141**, a storage **142**, and an image controller **143**, for example. The controller **141** includes an arithmetic unit such as a CPU. The storage **142** includes a memory, such as a ROM and a RAM. The control apparatus **14** may be composed of a single controller performing centralized control or may be composed of plural controllers that perform decentralized control in cooperation with each other. The controller **141** determines whether an action mode instruction inputted by the operation handles **11** is to be executed by the surgical instruments **40** or to be executed by the endoscope **50**, depending on the state of the operation pedal section **12**. When determining that the action mode instruction inputted by the operation handles **11** is to be executed by any one of the surgical instruments **40**, the controller **141** transmits the action mode instruction to the corresponding robot arm **21a** through the controller **24**. The robot arm **21a** is thereby driven by the controller **24** and thus movement of the surgical instrument **40** attached to the robot arm **21a** is controlled.

(49) When determining that the action mode instruction inputted by the operation handles **11** is to be executed by the endoscope **50**, the controller **141** transmits the action mode instruction to the robot arm **21b** through the controller **24**. The robot arm **21b** is thereby driven for control of movement of the endoscope **50** attached to the robot arm **21b**.

(50) The storage **142** stores control programs corresponding to the types of the surgical instrument **40**, for example. The controller **141** reads the stored control programs according to the types of the attached surgical instruments **40**. The action mode instructions from the operation handles **11** and/or the operation pedal section **12** of the remote control apparatus **10** thereby cause the respective surgical instruments **40** to perform proper movements.

(51) The image controller **143** transmits images acquired by the endoscope **50** to the display **13**.

The image controller **143** performs processing and modifying the images when needed.

(52) (Configurations of Surgical Instrument, Adaptor, Drape, and Robot Arm)

(53) With reference to FIGS. **4** to **6**, the configurations of the surgical instrument **40**, an adaptor **60**, a drape **70**, and the robot arm **21a** are described.

(54) Here, the direction in which the surgical instrument **40** (the direction in which the shaft **42** extends) is defined as the Y direction, the distal side (the side toward the end effector **43**) of the surgical instrument **40** along the Y direction is defined as the Y1 direction, and the opposite side of the Y1 direction is defined as the Y2 direction. The direction in which the surgical instrument **40** and the adaptor **60** are adjacent to each other is defined as a Z direction, the surgical instrument **40** side along the Z direction is defined as a Z1 direction, and the opposite side of the Z1 direction is defined as a Z2 direction. Further, the direction orthogonal to the Y direction and the Z direction is referred to as an X direction, one side along the X direction is referred to as an X1 direction, and the other side along the X direction is referred to as an X2 direction.

(55) As illustrated in FIGS. **4** and **5**, the surgical instrument **40** is detachably attached to the robot arm **21a** of the robotic surgical system **100**. Specifically, the surgical instrument **40** is detachably attached to the robot arm **21a** via the adaptor **60**. The adaptor **60** is a drape adaptor configured to sandwich a sterile drape **70** to cover the robot arm **21a**, in conjunction with the robot arm **21a**.

(56) The surgical instrument **40** is attached to the Z1 side of the adaptor **60**. The adaptor **60** is attached to the Z1 side of the robot arm **21a**.

(57) The robot arm **21a** is used in a clean area and is covered with the drape **70**. In operation rooms, clean technique is used in order to prevent surgical incision sites and medical equipment from being contaminated by pathogen, foreign matters, or the like. The clean technique defines a clean area and a contaminated area, which is outside the clean area. The surgery sites are located in the clean area. Members of the surgical team, including the operator, make sure that only sterile objects are placed in the clean area during surgery and perform sterilization for an object which is to be moved to the clean area from the contaminated area. Similarly, when an assistant, as one of the members of the surgical team including the operator, places their hands in the contaminated area, the member sterilize their hands before directly touching objects located in the clean area. Instruments used in the clean area are sterilized or are covered with the drapes **70** that are sterilized.

(58) As illustrated in FIG. **5**, the drape **70** includes a body section **71** that covers the robot arm **21a** and an attachment section **72** sandwiched between the robot arm **21a** and the adaptor **60**. The body section **71** is made of a flexible film member. The flexible film member is made of a resin material, such as thermoplastic polyurethane and polyethylene. The body section **71** includes an opening so that the robot arm **21a** is engaged with the adaptor **60**. To the opening of the body section **71**, the attachment section **72** is provided. The attachment section **72** is made of a resin mold member. The resin mold member is made of a resin member such as polyethylene terephthalate. The attachment section **72** is harder (less flexible) than the body section **71**. The attachment section **72** includes an opening so that the robot arm **21a** is engaged with the adaptor **60**. The opening of the attachment section **72** may be provided corresponding to the section where the robot arm **21a** is engaged with the adaptor **60**. The opening of the attachment section **72** may include plural openings corresponding to plural sections at which the robot arm **21a** is engaged with the adaptor **60**.

(59) As illustrated in FIGS. **5** and **6**, the surgical instrument **40** includes plural (four) driven members **44a**, **44b**, **44c**, and **44d**. The driven members **44a**, **44b**, **44c**, and **44d** are provided within the housing **41** and are rotatable about the respective rotation axes extending along the Z axis. The plural driven members **44a** to **44d** are provided to operate (drive) the end effector **43**. The driven members **44a**, **44b**, **44c**, and **44d** are connected to the end effector **43** via soft elongate elements W (such as wires or cables) passing through the inside of the shaft **42**. With this, rotations of the driven members **44b** to **44d** drive the elongate element W, which operate (drive) the end effector **43**. In addition, the driven member **44a** is connected to the shaft **42** through gears **42a** (see FIG. **7**). With this, the shaft **42** is rotated with rotation of the driven member **44a**, and the end effector **43** is

rotated with rotation of the shaft **42**.

(60) To transmit driving forces from the robot arm **21a** to the end effector **43**, each of the driven members **44a** to **44d** includes a projection **441** or **442**, which is engaged with the corresponding drive transmission member **61** of the adaptor **60**. Each of the projections **441** and **442** is projected from the Z2 side surface of the corresponding driven member **44a**, **44b**, **44c**, or **44d** toward the side of the adaptor **60** (the Z2 side). Each of the projections **441** and **442** includes plural projection portions that arranged in a straight line. The protrusions **441** provided to the driven members **44a** and **44b** have different shapes from that of the protrusions **442** provided to the driven members **44c** and **44d**.

(61) As illustrated in FIG. 5, the adaptor **60** includes a plurality (four) of the drive transmission members **61**. The drive transmission members **61** are configured to transmit the driving forces from the robot arm **21a** to the driven members **44a** to **44d** of the surgical instrument **40**. That is, the drive transmission members **61** are provided so as to correspond to the driven members **44a** to **44d** of the surgical instrument **40**. The drive transmission members **61** are rotatable about the respective rotation axes, which extend along the Z direction.

(62) Each of the drive transmission members **61** includes an engagement recess **611** which is engaged with the projection **441** or **442** of the corresponding driven members **44a**, **44b**, **44c**, and **44d** of the surgical instrument **40**. The engagement recess **611** is located at the surgical instrument **40** side (the Z1 side) of the drive transmission member **61** and is recessed from the Z1 side surface of the drive transmission member **61**, toward the Z2 direction, opposite to the surgical instrument **40**. Each of the drive transmission members **61** includes an engagement recess, which is provided on the Z2 side surface thereof and is configured to be engaged with a corresponding one of engagement projections **213** of the robot arm **21a**.

(63) The robot arm **21a** includes a frame **211**, plural (four) drive parts **212**, and the plural engagement projections **213**. The plural drive part **212** are provided corresponding to the plural (four) driven members **44a** to **44d** of the surgical instrument **40** and corresponding to the plural (four) drive transmission members **61** of the adaptor **60**. Each of the drive parts **212** includes an absolute encoder and a servomotor and is configured to drive the engagement projection **213** to rotate about the rotational axes thereof extending in the Z direction. The engagement projection **213** is engaged with the engagement recess provided on the Z2 side surface of the corresponding drive transmission member **61**. The engagement projection **213** is projected from the Z1 side surface of the robot arm **21a** toward the Z1 side (the adaptor **60** side). The drive parts **212** are configured to drive the drive transmission members **61** of the adaptor **60**, engaged with the engagement projections **213** of the drive parts **212**, to rotate about the rotational axes B of the drive transmission members **61** extending in the Z direction, so as to drive the driven members **44a** to **44d** of the surgical instrument **40**, engaged with the drive transmission members **61**, to rotate about the rotational axes of the driven members **44a** to **44d** extending in the Z direction.

(64) (Detailed Configuration of Surgical Instrument)

(65) With reference to FIGS. 7 to 9, the configuration of the surgical instrument **40** is described in detail below. Here, the case is described below in which the end effector **43** of the surgical instrument **40** is a grasping forceps including the pair of jaw members **43a** and **43b**.

(66) As illustrated in FIGS. 7 and 8, the elongate elements W are wound around the driven members **44b** to **44d** of the surgical instrument **40**. That is, the elongate elements W are connected to the driven members **44b** to **44d**.

(67) The elongate element W3 is wound around the driven member **44b** of the surgical instrument **40**. Specifically, a first portion W3a of the elongate element W3 is wound clockwise around an upper portion of the driven member **44b**, and a second portion W3b of the elongate element W3 is wound counterclockwise around a lower portion of the driven member **44b**.

(68) The elongate element W1 is wound around the driven member **44c** of the surgical instrument **40**. Specifically, a first portion W1a of the elongate element W1 is wound clockwise around an

upper portion of the driven member **44c**, and a second portion **W1b** of the elongate element **W1** is wound counterclockwise around a lower portion of the driven member **44c**.

(69) The elongate element **W2** is wound around the driven member **44d** of the surgical instrument **40**. Specifically, a first portion **W2a** of the elongate element **W2** is wound clockwise around an upper portion of the driven member **44d**, and a second portion **W2b** of the elongate element **W2** is wound counterclockwise around a lower portion of the driven member **44d**.

(70) The elongate elements **W** extend from the driven members **44b** to **44d** through the shaft **42** to the end effector **43**, are wound around the end effector **43**, return to the driven members **44b** to **44d** through the shaft **42**. Further, the elongate elements **W** are wound around built-in pulleys **45** in the housing **41**, respectively. The built-in pulleys **45** are retained by a pulley retainer **451**.

(71) As illustrated in FIGS. **8** and **9**, when the driven member **44c** rotates about the rotary axis thereof, the rotation of the driven member **44c** operates the jaw member **43a**, which is one of the pair of jaw members **43a** and **43b** of the end effector **43**. Specifically, the driven member **44c** is rotated by the drive part **212** to drive the elongate element **W1**. The elongate element **W1** extends through the inside of the shaft **42** and connects the jaw member **43a** and the driven member **44c**. When the driven member **44c** is rotated in the C1 direction (see FIG. **8**), the first portion **W1a** of the elongate element **W1** is pulled and the second portion **W1b** of the elongate element **W1** is fed, so as to drive the jaw member **43a** to move in the C1a direction (see FIGS. **9A** and **9B**), which is a direction to open the jaw member **43a**. When the driven member **44c** is rotated in the C2 direction (see FIG. **8**) opposite to the C1 direction, the first portion **W1a** of the elongate element **W1** is fed and the second portion **W1b** of the elongate element **W1** is pulled, so as to drive the jaw member **43a** to move in the C2a direction (see FIGS. **9A** and **9B**), which is a direction to close the jaw member **43a**.

(72) When the driven member **44d** rotates about the rotary axis thereof, the rotation of the driven member **44d** operates the jaw member **43b**, which is one of the pair of jaw members **43a** and **43b** of the end effector **43**. Specifically, the driven member **44d** is rotated by the drive part **212** to drive the elongate element **W2**. The elongate element **W2** extends through the inside of the shaft **42** and connects the jaw member **43b** and the driven member **44d**. When the driven member **44d** is rotated in the C3 direction (see FIG. **8**), the first portion **W2a** of the elongate element **W2** is pulled and the second portion **W2b** of the elongate element **W2** is fed, so as to drive the jaw member **43b** to move in the C3a direction (see FIGS. **9A** and **9B**), which is a direction to open the jaw member **43b**.

When the driven member **44d** is rotated in the C4 direction (see FIG. **8**) opposite to the C3 direction, the first portion **W2a** of the elongate element **W2** is fed and the second portion **W2b** of the elongate element **W2** is pulled, so as to drive the jaw member **43b** to move in the C4a direction (see FIGS. **9A** and **9B**), which is a direction to close the jaw member **43b**. That is, the pair of jaw members **43a** and **43b** are opened and closed with respect to each other by the movements of the elongate elements **W** driven by the driven members **44b** and **44c**.

(73) By being rotated about the rotation axis thereof, the driven member **44b** operates a wrist portion **43c** of the end effector **43**. Specifically, the driven member **44b** is rotated by the drive part **212** to drive the elongate element **W3**. The elongate element **W3** extends through the inside of the shaft **42** and connects the wrist portion **43c** and the driven member **44b**. When the driven member **44b** is rotated in the C5 direction (see FIG. **8**), the first portion **W3a** of the elongate element **W3** is pulled and the second portion **W3b** of the elongate element **W3** is advanced, so as to drive the wrist portion **43c** to move in the C5a direction (see FIGS. **9A** and **9B**). When the driven member **44d** is rotated in the C6 direction (see FIG. **8**) opposite to the C5 direction, the second portion **W3b** of the elongate element **W3** is pulled and the first portion **W3a** of the elongate element **W3** is fed, so as to drive the wrist portion **43c** to move in the C6a direction (see FIGS. **9A** and **9B**), which is opposite to the C5a direction.

(74) When the drive part **212** rotates the driven member **44a** about the rotation axis thereof with the gear portion **443** of the driven member **44a** being engaged with a gear portion **42a** connected to the

proximal end of the shaft **42**, the shaft **42** is driven to operate the end effector **43**. Specifically, when the driven member **44a** rotates in the C7 direction (see FIG. **8**), the shaft **42** is driven to rotate in the C7a direction (see FIGS. **9A** and **9B**) and thus the end effector **43** is driven to rotate in the C7a. When the driven member **44a** rotates in the C8 direction (see FIG. **8**), the shaft **42** is driven to rotate in the C8a direction (see FIGS. **9A** and **9B**), which is opposite to the C7a direction, and thus the end effector **43** is driven to rotate in the C8a direction.

(75) (Control of Opening and Closing Pair of Jaw Members)

(76) Next, a control of opening and closing the pair of jaw members **43a** and **43b** is described with reference to FIGS. **11** to **14**. In the following, the drive part **212** of the robot arm **21a** that drives the jaw member **43a** is referred to as a drive part **212a**, and the drive part **212** of the robot arm **21a** that drives the jaw member **43b** is referred to as a drive part **212b**.

(77) As illustrated in FIGS. **11** and **12**, the controller **24** included in the patient-side apparatus **20** is configured, based on a command jaw opening angle (or a command angle) for the opening angle θ of the pair of jaw members **43a** and **43b**, to drive the motors of the drive part **212a** and **212b** to rotate. The command jaw opening angle is a command value of the opening angle θ between the pair of jaw members **43a** and **43b**, and is associated with (is generated based on) the input of the operation handle **11**. The command jaw opening angle is converted by the controller **24** into a command value (a drive signal) to rotate the motors of the drive parts **212a** and **212b** by rotation amounts thereof so as to make the opening angle θ between the pair of jaw members **43a** and **43b** corresponded to the opening angle between the pair of grip members **11b** of the operation handle **11**.

(78) FIG. **12** is a graph illustrating a relationship between the opening angle between the pair of grip members **11b** of the operation handle **11** and the command jaw opening angle for the opening angle between the pair of jaw members **43a** and **43b** in a mode other than a restriction mode described later). In the graph illustrated in FIG. **12** (and FIG. **13**), the horizontal axis indicates the opening angle between the pair of grip members **11b**, and the vertical axis indicates the command jaw opening angle for the opening angle between the pair of jaw members **43a** and **43b**. The graph of FIG. **12** indicates that, as the opening angle between the pair of grip members **11b** becomes smaller, the pair of jaw members **43a** and **43b** gradually close. The graph of FIG. **12** also indicates that when the motors of the drive parts **212a** and **212b** are energized to try to rotate the pair of jaw members **43a** and **43b** in the closing direction thereof after the pair of jaw members **43a**, **43b** are completely closed (that is, after the opening angle θ becomes zero), a tightening force (a grasping force) is generated between the pair of jaw members **43a** and **43b**.

(79) Here, regardless of whether or not the object T (see FIG. **14**) exists between the pair of jaw members **43a** and **43b**, if the pair of jaw members **43a** and **43b** can be tightened up to a minimum value of the command jaw opening angle illustrated in the graph of FIG. **12**, an excessive grasping force may occur.

(80) In light of this, in a first embodiment, as illustrated in FIGS. **13** and **14**, the controller **24** is configured to monitor current values (hereinafter, motor current values) applied to the motors of the drive parts **212a** and **212b** during the closing operation of the pair of jaw members **43a** and **43b**, and to, when it is determined based on the motor current values that the object T is being grasped, drive the drive parts **212a** and **212b** in the restriction mode in which a magnitude of the command jaw opening angle is restricted. That is, the controller **24** is configured to determine whether or not the motor current value of the drive part **212a** or **212b** exceeds a predetermined threshold value during the closing operation of the pair of jaw members **43a** and **43b**, and when it is determined that the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value, to drive the drive parts **212a** and **212b** in the restriction mode, in which the magnitude of the command jaw opening angle is restricted. Accordingly, when it is determined that the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value during the closing operation of the pair of jaw members **43a** and **43b**, the drive parts **212a** and **212b** are driven in the

restriction mode in which the magnitude of the command jaw opening angle is restricted.

Therefore, it is possible to prevent the grasping force from becoming excessive. As a result, an appropriate grasping force can be obtained. Further, since the magnitude of the command jaw opening angle is restricted, it is not necessary to provide load cells or torque sensors to the elongate elements W. Therefore, it is possible to make the structure around the elongate elements W less complicated than a case where the load cells or the torque sensors are provided to the elongate elements W. As a result, it is possible to achieve an appropriate grasping force while suppressing the structure around the elongate elements W from being complicated.

(81) Specifically, the controller **24** is configured, upon determining that the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value during the closing operation of the pair of jaw members **43a** and **43b**, to drive the drive parts **212a** and **212b** in the restriction mode in which a minimum value V2 (a lower limit V2) of the command jaw opening angle in the restriction mode is set (restricted) greater than an initial value, which is a minimum value V4 (a lower limit V4) of the command jaw opening angle in a mode other than the restriction mode, in such a manner that a difference between the minimum value V2 of the command jaw opening angle in the restriction mode and the value V1 of the command jaw opening angle when the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value is a predetermined value V3. That is, the drive parts **212a** and **212b** are driven in the restriction mode in which the minimum value V2 of the command jaw opening angle is set (restricted) greater than the minimum value V4 of the command jaw opening angle in the mode other than the restriction mode. Therefore, it is possible to easily prevent the grasping force from becoming excessive. As a result, an appropriate grasping force can be achieved more easily.

(82) In other words, the controller **24** is configured, in response to determining that the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value, to reset the minimum value (the initial value V4) of the command jaw opening angle to the value (e.g., V2) greater than the initial value V4, in such a manner that a closing angle of the jaw members **43a** and **43b** from a time when it is determined that the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value is set to the predetermined value V3 (constant value).

(83) In the restriction mode, when it is determined that the value of the command jaw opening angle generated corresponding to the opening angle between the pair of grip members **11b** of the operation handle **11** is less than the minimum value V2 of the command jaw opening angle in the restriction mode (the restricted minimum value V2), the controller **24** is configured to drive the drive parts **212a** and **212b** in accordance with the command jaw opening angle that is set (restricted) to the minimum value V2 of the command jaw opening angle in the restriction mode (the restricted minimum value V2). Further, in the restriction mode, when it is determined that the value of the command jaw opening angle generated corresponding to the opening angle between the pair of grip members **11b** of the operation handle **11** is equal to or greater than the minimum value V2 of the command jaw opening angle in the restriction mode (the restricted minimum value V2), the controller **24** is configured to drive the drive parts **212a** and **212b** in response to the value of the command jaw opening angle generated corresponding to the opening angle between the pair of grip members **11b** of the operation handle **11** in a same manner as in the mode (FIG. 12) other than the restriction mode.

(84) Further, in a first embodiment, the predetermined value V3, which is the difference between the value V1 of the command jaw opening angle when the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value and the minimum value V2 of command jaw opening angle in the restriction mode, is set to the same as a difference V6 between the value V5 of the command jaw opening angle when the opening angle θ of the pair of jaw members **43a** and **43b** is zero and the initial minimum value V4, which is the minimum value V4 of the command jaw opening angle in the mode other than the restriction mode. As a result, the drive parts **212a** and **212b** can be driven in the restriction mode in which the minimum value of the command jaw

opening angle is restricted in such a manner that the difference between the value **V1** of the command jaw opening angle when the motor current values of the drive part **212a** or **212b** exceeds the predetermined threshold value and the minimum value **V2** of the command jaw opening angle in the restriction mode is set to the same as the difference **V6** between the value **V5** of the command jaw opening angle when the opening angle θ of the pair of jaw members **43a** and **43b** is zero and the initial minimum value **V4** of the command jaw opening angle. As a result, in the restriction mode, it is possible to generate a constant grasping force substantially corresponding to a maximum grasping force when the pair of jaw members **43a** and **43b** do not grasp the object **T**. Therefore, it is possible to more easily prevent an excessive grasping force from being generated. As a result, an appropriate grasping force can be achieved more easily.

(85) Further, in a first embodiment, the controller **24** is configured to switch to the restriction mode, when it is determined that the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value, and the value of the command jaw opening angle generated is smaller than the previous value of the command jaw opening angle (the value immediately before), which is stored in the memory **25** in the patient-side apparatus **20** (see FIG. 1). The memory **25** stores therein the value of the command jaw opening angle at predetermined time intervals.

Accordingly, when it is detected, based on the motor current values of the drive parts **212a** and **212b**, that the pair of jaw members **43a** and **43b** are grasping the object **T** and it is detected, based on the value of the command jaw opening angle generated becoming smaller than the previous value of the command jaw opening angle, that the pair of jaw members **43a** and **43b** are closing, the controller switches to restriction mode. As a result, it is possible to switch to the restriction mode at an appropriate timing when the grasping force of the pair of jaw members **43a** and **43b** should be restricted. Note that the memory **25** is an example of a storage or a storage device.

(86) Specifically, the controller **24** is configured to switch to the restriction mode, when it is repeatedly determined multiple times (for example, two times) that the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value and the value of the command jaw opening angle generated gets smaller than the previous value of the command jaw opening angle stored immediately before in the memory **25**. Accordingly, the controller **24** can switch to the restriction mode, when it is detected with certainty that the pair of jaw members **43a** and **43b** are grasping the object **T** based on the motor current values of the drive parts **212a** and **212b** and that the pair of jaw members **43a** and **43b** are closing based on the value of the command jaw opening angle generated getting smaller than the previous value of the command jaw opening angle. As a result, it is possible to switch to the restriction mode at a more appropriate timing.

(87) More specifically, the controller **24** is configured to switch to the restriction mode, when it is repeatedly determined multiple times that one of the motor current value of the drive part **212a** and the motor current value of the drive part **212b** exceeds the predetermined threshold value and that the value of the command jaw opening angle gets smaller than the previous value of the command jaw opening angle stored immediately before in the memory **25**.

(88) Further, in a first embodiment, the controller **24** is configured, when the motor current value of the drive part **212a** or **212b** gets equal to or less than the predetermined threshold value, to release (finish) the restriction mode. Thereby, when it is detected based on the motor current values of the drive parts **212a** and **212b** that the grasping of the object **T** by the pair of jaw members **43a** and **43b** is released, the restriction mode is released (ended). As a result, the restriction mode can be released at an appropriate timing when the restriction mode should be released.

(89) Specifically, in a state where the surgical instrument **40** is operable (is allowed to be operated) by the operation handle **11**, the controller **24** is configured to release the restriction mode, when it is detected that the motor current value of the drive part **212a** or **212b** gets equal to or less than the predetermined threshold value and it is detected that the value of the command jaw opening angle generated exceeds the value of the command jaw opening angle at the start of the restriction mode (that is, **V1**) or a value close to the maximum value of the command jaw opening angle (for

example, a value one degree smaller than the maximum value). Further, in the state where the surgical instrument **40** is operable by the operation handle **11**, the controller **24** can release the restriction mode, when it is detected, based on the motor current value of the drive part **212a** or **212b**, that the grasping of the object T by the pair of jaw members **43a** and **43b** is released and it is detected, based on the value of the command jaw opening angle exceeding the value of the command jaw opening angle at the start of the restriction mode, that the pair of jaw members **43a** and **43b** are opened more than the jaw opening angle at the start of the restriction mode. As a result, the restriction mode can be released (ended) at a more appropriate timing. Further, when it is detected, based on the motor current values of the drive parts **212a** and **212b**, that grasping of the object T by the pair of jaw members **43a** and **43b** is released and it is detected, based on the value of the command jaw opening angle exceeding the value close to the maximum value of the command jaw opening angle (an approximate maximum value of the command jaw opening angle or a value slightly smaller than the maximum value), that the pair of jaw members **43a** and **43b** are in a maximum open state, the controller can release the restriction mode. As a result, even in a case where the pair of jaw members **43a** and **43b** have been opened to the maximum at the start of the restriction mode and thus the pair of jaw members **43a** and **43b** cannot be opened more from the state at the start of the restriction mode, the restriction mode can be released.

(90) Note that the controller **24** is configured, when the head sensor **13a** (see FIG. 1) detects the presence of the operator's head and the operator performs a predetermined operation using the operation handle **11**, to determine (detect) that the surgical instrument **40** is in an operable state (hereinafter, may be referred to as a following state) in which the surgical instrument **40** can be operated by the operation handle **11**. Further, the controller **24** is configured, when the head sensor **13a** detects no presence of the operator's head, to determine (detect) that the surgical instrument **40** is in an inoperable state (hereinafter, may be referred to as an out-of-following state or a no-following state) in which the surgical instrument **40** is not allowed to be operated by the operation handle **11**.

(91) Further, in a first embodiment, in the out-of-following state, the controller **24** is configured, when it is detected that the motor current value of the drive part **212a**, **212b** gets equal to or less than the predetermined threshold value, to release (end) the restriction mode regardless of the value of the command jaw opening angle, with considering that the command jaw opening angle does not change in the out-of-following state. Accordingly, when it is detected, based on the motor current value of the drive part **212a**, **212b**, that grasping of the object by the pair of jaw members **43a** and **43b** is released in the out-of-following state, the restriction mode can be released regardless of the value of the command jaw opening angle. As a result, the restriction mode can be released under the condition suitable for the out-of-following state. Note that after starting the restriction mode in the following state, the following state may be switched to the out-of-following state and the restriction mode may be continued in the out-of-following state. In this case, if the above conditions are satisfied, the restriction mode is released.

(92) Further, in a first embodiment, the predetermined threshold value is set based on frictional forces of the motors of the drive parts **212a** and **212b**. With this, the predetermined threshold value for detecting whether the pair of jaw members **43a** and **43b** grasp an object T can be set in consideration of the influence of the current values generated due to the frictional forces of the drive parts **212a** and **212b**. As a result, it is possible to accurately detect based on the predetermined threshold value whether the pair of jaw members **43a** and **43b** is grasping an object T, as compared with a case where the influence of the current values generated due to the frictional forces of the drive parts **212a** and **212b** is not taken into consideration.

(93) Specifically, the predetermined threshold value may be set by a following equation (1). As expressed in the equation (1) below, the predetermined threshold value is set in consideration of a dynamic friction of the motor of the drive part **212a** (**212b**) and a product of a viscous friction of the motor of the drive part **212a** (**212b**) and an axial speed (rotational speed) of the drive part **212a**

(212b), with respect to a reference value. Since the current value that is generated due to the viscous friction of the drive part 212a (212b) varies according to the axial speed of the drive part 212a (212b), the current value is considered as a value that varies according to the axial speed of the drive part 212a (212b) in the equation (1). Further, since the current value that is generated due to the dynamic friction of the drive part 212a (212b) is constant regardless of the axial speed of the drive part 212a (212b), the current value is considered as a constant value that does not vary according to the axial speed of the drive part 212a (212b) in the equation (1).

$\alpha = \alpha_1 + (a \times A_s + b)$ (1) α : the predetermined threshold value α_1 : the reference value a : a constant related to the viscous friction of the drive part b : a constant related to the dynamic friction of the drive part A_s : a variable of the axial speed (rotational speed) of the drive part

(94) “ α_1 ” is a constant that serves as a reference for the predetermined threshold value. “ a ” is a constant related to the viscous friction generated due to grease or the like in a speed reducer or the like of the drive part 212a (212b). “ b ” is a constant related to the dynamic friction generated due to the rotation in the speed reducer or the like of the drive part 212a (212b). The constants “ α_1 ”, “ a ” and “ b ” have been obtained in advance by experiments and the like. The constants “ α_1 ”, “ a ” and “ b ” are not particularly limited. However, as an example, “ α_1 ” is 0.2 (Arms), “ a ” is 0.0006 (Arms/sec/deg), and “ b ” is 0.1194 (Arms). “ A_s ” is a variable representing the axial speed (rotational speed) of the drive part 212a (212b). “ A_s ” can be acquired based on an output of the position detector such as the encoder of the drive part 212a (212b).

(95) The controller 24 is configured, when determining whether or not to switch to the restriction mode, to compare the predetermined threshold value of the drive part 212a that is obtained on the real time based on the equation (1) with the motor current value of the drive part 212a that is obtained on the real time from the current detector 220a that is provided for the drive part 212a and configured to detect the motor current value of the drive part 212a. Likewise, the controller 24 is configured, when determining whether or not to switch to the restriction mode, to compare the predetermined threshold value of the drive part 212b that is obtained on the real time based on the equation (1) with the motor current value of the drive part 212b that is obtained on the real time from the current detector 220b that is provided for the drive part 212b and is configured to detect the motor current value of the drive part 212b.

(96) Further, in a first embodiment, as illustrated in FIGS. 9 and 10, the end effector 43 includes a pulley 43d around which the elongate element W1 for driving the jaw member 43a is wound, and a pulley 43e around which the elongate element W2 for driving the jaw member 43b is wound. The pulley 43d is provided with an engagement hole 43da that engages with an engagement pin 43aa of the jaw member 43a, and the pulley 43e is provided with an engagement hole 43ea that engages with an engagement pin 43ba of the jaw member 43b. The pair of jaw members 43a and 43b are configured, when the pulleys 43d and 43e are rotated, to open and close with respect to each other about a shaft 43g (including a rotational axis) different from a shaft 43f (including a rotational axis) of the pulleys 43d and 43e. That is, the pair of jaw members 43a and 43b and the pulleys 43d and 43e form a link mechanism. As a result, it is possible to generate a large grasping force by the pair of jaw members 43a and 43b. Further, in the case where the pair of jaw members 43a and 43b are used that have a large grasping force, it is possible to suppress generation of an excessive grasping force by using the restriction mode described above. Note that the engagement hole 43da and the engagement hole 43ea are examples of engagement portions for driving the jaw member.

(97) The elongate element W1 is wound around the pulley 43d, so that the pulley 43d is rotated by movements of the elongate element W1. The elongate element W2 is wound around the pulley 43e, so that the pulley 43e is rotated by movements of the elongate element W2. The pulley 43d and 43e are rotatably supported by the common shaft 43f. The pair of jaw members 43a and 43b are rotatably supported by the common shaft 43g. The shaft 43f and the shaft 43g extend substantially parallel to each other. The shaft 43g is arranged between the distal end side portion and the proximal end side portion of the pair of jaw members 43a and 43b. The shaft 43g is arranged

between the pulleys **43d** and **43e** and the distal end side portion of the pair of jaw members **43a** and **43b**. The proximal end side portion of the jaw member **43a** is provided with the engagement pin **43aa** to be inserted into the engagement hole **43da** of the pulley **43d**. The proximal end side portion of the jaw member **43b** is provided with the engagement pin **43ba** to be inserted into the engagement hole **43ea** of the pulley **43e**.

(98) When the pulley **43d** is rotated around the shaft **43f** by the elongate element **W1**, the rotational driving force is transmitted to the jaw member **43a** via the engagement pin **43aa** engaged with the engagement hole **43da**. With this, the jaw member **43a** is rotated about the shaft **43g**. At this time, the rotational driving force of the pulley **43d** is amplified by the action of the link mechanism and transmitted to a grasping portion of the jaw member **43a** provided at the distal end side portion of the jaw member **43a**. Similarly, when the pulley **43e** is rotated around the shaft **43g** by the elongate element **W2**, the rotational driving force is transmitted to the jaw member **43b** via the engagement pin **43ba** engaged with the engagement hole **43ea**. With this, the jaw member **43b** is rotated around the shaft **43g**. At this time, the rotational driving force of the pulley **43e** is amplified by the action of the link mechanism and transmitted to a grasping portion of the jaw member **43b** provided at the distal end side portion of the jaw member **43b**.

(99) Here, in a case where a relatively large object **T** (see FIG. **14**) is grasped by the pair of jaw members **43a** and **43b** that can generate the large grasping force as illustrated in FIGS. **9A** and **9B**, an excessive grasping force is likely to be generated. If the excessive grasping force is generated, the elongate elements **W1** and **W2** may fall off from the built-in pulleys **45** (see FIG. **8**). That is, when the excessive grasping force is generated, for example, one of the first portion **W1a** and the second portion **W1b** of the elongated element **W1** that is fed out is excessively fed out, so that the elongate element **W1** may be loosened around the built-in pulley **45** and may be fall off from the built-in pulley **45**. Note that the same may happen for the elongate element **W2**.

(100) Therefore, in a first embodiment, as illustrated in FIG. **15**, a fall-off prevention part **46** configured to prevent the elongate elements **W1** and **W2** from falling off from the built-in pulleys **45** is provided in the vicinity of the built-in pulleys **45**. The fall-off prevention part **46** includes recesses **461** configured to retain the elongate elements **W1** and **W2** between the recesses **461** and groove portions of the built-in pulleys **45**. The recesses **461** cover the groove portions of the built-in pulleys **45** from the side, so that the elongate elements **W1** and **W2** positioned in the groove portions of the built-in pulleys **45** are retained in the spaces between the groove portions of the built-in pulleys **45** and the recesses **461**. Further, the number of the recesses **461** is four corresponding to a total of four portions of the elongate elements **W1** and **W2**, which includes the first portion **W1a** and the second portion **W1b** of the elongate element **W1** and the first portion **W2a** and the second portion **W2b** of the elongate element **W2**.

(101) (Control Process of Starting Restriction Mode)

(102) Next, with reference to the flowchart of FIG. **16**, a control process for starting the restriction mode by the controller **24** in the following state is described below.

(103) As illustrated in FIG. **16**, first, in step **S1**, it is determined whether or not the motor current value of the drive part **212a**, **212b** exceeds the predetermined threshold value. That is, in step **S1**, it is determined whether or not the motor current value of the drive part **212a**, **212b** has increased due to the pair of jaw members **43a** and **43b** grasping the object **T**.

(104) Specifically, in step **S1**, it is determined whether or not the motor current values of the drive part **212a** exceeds the predetermined threshold value of the drive part **212a** according to the above equation (1), or the motor current value of the drive part **212b** exceeds the predetermined threshold value according to the above equation (1). If either the motor current value of the drive part **212a** or the motor current value of the drive part **212b** do not exceed the predetermined threshold value, the process of step **S1** is repeated. If at least one of the motor current value of the drive part **212a** and the motor current value of the drive part **212b** exceeds the predetermined threshold value, the process proceeds to step **S2**.

(105) Then, in step S2, it is determined whether or not the value of the command jaw opening angle generated corresponding to the opening angle between the pair of grip members **11b** of the operation handle **11** is smaller than the previous value of the command jaw opening angle which is stored immediately before. If the value of the command jaw opening angle generated is not smaller than the previous value of the command jaw opening angle, the process proceeds to step S1. If the value of the command jaw opening angle generated is smaller than the previous value of the command jaw opening angle, the process proceeds to step S3. Note that, for example, upon proceeding from steps S1, S2, and S3 to step S1, the value of the command jaw opening angle is stored in the memory **25** as the previous value of the command jaw opening angle.

(106) Then, in step S3, it is determined whether or not the conditions of steps S1 and S2 are satisfied twice in a row. If the conditions of steps S1 and S2 are not satisfied twice in a row, the process proceeds to step S1. If the conditions of steps S1 and S2 are satisfied twice in a row, the process proceeds to step S4.

(107) Then, in step S4, the mode is switched to the restriction mode, and the value of the command jaw opening angle when the restriction mode is started is stored in the memory **25**. In the restriction mode, the magnitude of the command jaw opening angle is restricted as described above. After that, the restriction mode is continued until the restriction mode is released (ended).

(108) (Control Process of Releasing Restriction Mode)

(109) Next, with reference to the flowchart of FIG. **17**, a control process for releasing the restriction mode by the controller **24** in the following state is described below.

(110) As illustrated in FIG. **17**, first, in step S11, it is determined whether or not the motor current values of the drive parts **212a** and **212b** are equal to or less than the predetermined threshold value. That is, in step S11, it is determined whether or not the motor current value of the drive part **212a**, **212b** has decreased because the pair of jaw members **43a** and **43b** no longer grasp the object T.

(111) Specifically, in step S11, it is determined whether or not the motor current value of the drive part **212a** gets equal to or less than the predetermined threshold value of the drive part **212a** according to the above described equation (1), and the motor current value of the drive part **212b** gets equal to or less than the predetermined threshold value for the drive part **212b** according to the above described equation (1). If it is determined that both the motor current value of the drive part **212a** and the motor current value of the drive part **212b** are not equal to or less than the predetermined threshold value, the process of step S11 is repeated. If it is determined that both the motor current value of the drive part **212a** and the motor current value of the drive part **212b** are equal to or less than the predetermined threshold value, the process proceeds to step S12.

(112) Then, in step S12, it is determined whether or not the value of the command jaw opening angle generated corresponding to the opening angle between the pair of grip members **11b** of the operation handle **11** exceeds the value of the command jaw opening angle at the time of starting the restriction mode. If the value of the command jaw opening angle generated exceeds the value of the command jaw opening angle at the time of starting the restriction mode, the process proceeds to step S14. Then, in step S14, the restriction mode is released (ended).

(113) Further, in step S12, if the value of the command jaw opening angle generated does not exceed the value of the command jaw opening angle at the time of starting the restriction mode, the process proceeds to step S13.

(114) Then, in step S13, it is determined whether or not the value of the command jaw opening angle generated exceeds the value close to the maximum value of the command jaw opening angle (the approximate maximum value, e.g., a value one degree smaller than the maximum value). If the value of the command jaw opening angle generated does not exceed the value that is less than the approximate maximum value (e.g., the value one degree smaller than the maximum value), the process proceeds to step S11. If the value of the command jaw opening angle exceeds the approximate maximum value (e.g., the value one degree smaller than the maximum value), the process proceeds to step S14. Then, in step S14, the restriction mode is released (ended).

(115) (Control Process of Releasing Restriction Mode)

(116) Next, with reference to the flowchart of FIG. 18, the control process for releasing the restriction mode by the controller 24 in the out-of-following state is described below.

(117) As illustrated in FIG. 18, first, in step S21, it is determined whether or not the motor current value of the drive part 212a, 212b gets equal to or less than the predetermined threshold value. Note that the process of step S21 is the same as in step S11 illustrated in FIG. 17. That is, if it is determined that both the motor current value of the drive part 212a and the motor current value of the drive part 212b are not equal to or less than the predetermined threshold value, the process of step S21 is repeated. If it is determined that both the motor current value of the drive part 212a and the motor current value of the drive part 212b are equal to or less than the predetermined threshold value, the process proceeds to step S22. Then, in step S22, the restriction mode is released (ended). Unlike the control process for releasing the restriction mode during the following state illustrated in FIG. 17, in the control process for releasing the restriction mode during the out-of-following state illustrated in FIG. 18, the conditions (steps S12 and S13) of the command jaw opening angle do not determined.

Second Embodiment

(118) Next, a second embodiment is described with reference to FIGS. 19 to 21. Unlike a first embodiment, in a restriction mode according to a second embodiment, the minimum value of the commanded jaw opening angle is limited to zero or less. The configurations in a second embodiment same as those of a first embodiment are designated by the same reference numerals in the drawings, and the description thereof may be omitted for avoid redundancy.

(119) (Configuration of Robotic Surgical System)

(120) As illustrated in FIG. 19, a robotic surgical system 300 according to a second embodiment includes a controller 324 instead of the controller 24 of a first embodiment. Note that the controller 324 is an example of a controller or a control device.

(121) FIG. 20 is a graph illustrating a relationship (indicated by a dash-dotted line) between a detected angle and the minimum value of the jaw command opening angle in addition to the relationship (the graph of FIG. 12) between the opening angle between the pair of grip members 11b of the operation handle 11 and the command jaw opening angle for the opening angle between the pair of jaw members 43a and 43b. In the graph illustrated in FIG. 20, the horizontal axis indicates the opening angle between the pair of grip members 11b, and the vertical axis indicates the command jaw opening angle for the opening angle between the pair of jaw members 43a and 43b. Note that the detected angle is the opening angle θ of the pair of jaw members 43a and 43b when the motor current value of the drive part 212a, 212b exceeds the predetermined threshold value. For example, the detected angle can be estimated from the rotation angle of the motor of the drive part 212a, 212b detected by the controller 324 when the motor current value of the drive part 212a, 212b exceeds the predetermined threshold value, with taking into account the elongation of the elongate element W.

(122) FIG. 21 is a graph illustrating the relationship between the detected angle and the minimum value of the command jaw opening angle. In the graph illustrated in FIG. 21, the horizontal axis indicates the detected angle and the vertical axis indicates the minimum value of the command jaw opening angle for the opening angle between the pair of jaw members 43a and 43b. Further, in the graph of FIG. 21, the solid line indicates the relationship between the detected angle and the minimum value of the command jaw opening angle in the case where the control according to a first embodiment is performed, whereas the dot-dash line indicates the relationship between the detected angle and the minimum value of the commanded jaw opening angle in the case where the control according to a second embodiment is performed.

(123) In a second embodiment, as illustrated in FIGS. 20 and 21, the controller 324 is configured, when it is determined that the motor current values of the drive parts 212a and 212b during the closing operation of the pair of jaw members 43a and 43b excesses the predetermined threshold

value, drive the drive parts **212a** and **212b** in a restriction mode in which the minimum value **V2** of the command jaw opening angle is set (restricted) to a value that is greater than the initial value **V4** and is equal or less than zero. That is, in the restriction mode according to a second embodiment, not only the lower limit of the minimum value of the command jaw opening angle is restricted, but also the upper limit of the minimum value of the command jaw opening angle is restricted. Here, for example, in a case where the object **T** is a tissue and is pulled while being gripped by the pair of jaw members **43a** and **43b**, the thickness of the object **T** would become thinner. In a first embodiment described above, the minimum value of the command jaw opening angle for the pair of jaw members **43a** and **43b** may be greater than zero (see FIG. 21), and thus it may be possible that the object **T** may fall off the pair of jaw members **43a** and **43b** due to change in the thickness of the object **T**. To the contrary, in a second embodiment, even in the restriction mode, the minimum value of the commanded jaw opening angle is set to always be zero or less. Thus, even if the thickness of the object **T** changes, the pair of jaw members **43a** and **43b** is closed to follow the change in the thickness of the object **T**, so that the object **T** can be reliably grasped.

(124) Further, in a second embodiment, the upper limit **V11** of the minimum value **V2** of the commanded jaw opening angle in the restriction mode is set. The upper limit value **V11** is a negative value smaller than zero by a predetermined amount. Therefore, unlike a case where the upper limit value **V11** is zero, even if a backlash between the jaw member **43a** (**43b**) and the pulley **43d** (**43e**) exists, the pair of jaw members **43a** and **43b** can be completely closed in the restricted mode. The upper limit value **V11** is set to the negative value smaller by the predetermined amount corresponding to the backlash between the jaw member **43a** (**43b**) and the pulley **43d** (**43e**). For example, in a case where the backlash is 1.5 degrees, the upper limit value **V11** is set to -1.5 degrees.

(125) Further, in a second embodiment, the controller **324** is configured to, when it is determined that the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value during the closing operation of the pair of jaw members **43a** and **43b**, detect the rotation angle of the motor at the time when the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value, and to control, based on the detected rotation angle of the motor, whether or not to switch to the restriction mode (switching to the restriction mode). Specifically, the controller **24** is configured to, when it is determined that the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value during the closing operation of the pair of jaw members **43a** and **43b**, determine whether or not the detected angle is equal to or greater than a predetermined value **V12**, and to switch to the restriction mode when the detected angle is equal to or greater than the predetermined value **V12** and not to switch to the restriction mode when the detected angle is less than the predetermined value **V12**. As a result, it is possible to prevent unnecessary switching to the restriction mode in a range where the detected angle is small. When not switching to the restriction mode, the minimum value of the commanded jaw opening angle is the initial value **V4**.

(126) Further, the controller **324** is configured to, when it is determined that the motor current value of the drive part **212a** or **212b** exceeds the predetermined threshold value during the closing operation of the pair of jaw members **43a** and **43b** and the detected angle exceeds the predetermined value **V12**, switch to the restriction mode. Specifically, the controller **324** is configured, when the detected angle is equal to or greater than the predetermined value **V12** and less than a predetermined value **V13** (**V13**>**V12**), to switch to the restriction mode in which the minimum value **V2** of the command jaw opening angle is restricted to greater than the initial value **V4** and less than the upper limit value **V11**. In this case, the controller **324** is configured to restrict the minimum value **V2** of the commanded jaw opening angle so that the minimum value **V2** of the commanded jaw opening angle gradually decreases as the detected angle decreases. For example, the controller **324** is configured to obtain the minimum value **V2** of the commanded jaw opening angle according to Equation (2) below.

$V2=V4 \times R + V4 - c$ (2) V2: the minimum value of the command jaw opening angle in the restriction mode V4: the initial value (the minimum value of the command jaw opening angle before the restriction mode) R: a reduction ratio with respect to the detected angle (a variable that varies according to the detected angle) c: a predetermined constant

(127) Further, the controller **324** is configured, when the detected angle is equal to or greater than the predetermined value **V13**, to switch to the restriction mode in which the minimum value **V2** of the commanded jaw opening angle is limited to the upper limit value **V11**. In this case, the controller **324** is configured to limit the minimum value **V2** of the commanded jaw opening angle so that the minimum value **V2** of the commanded jaw opening angle is set to the upper limit value **V11** regardless of the value of the detected angle.

Modifications

(128) Note that one or more embodiments disclosed herein should be considered as exemplary in all respects and do not limit the invention. The scope of the invention is indicated by claims, not by explanation of one or more embodiments described above, and includes equivalents to the claims and all alterations (modification) within the same.

(129) In first and second embodiments described above, the case has been described in which the pair of jaw members are driven by the soft elongate elements formed of wires or cables. However, the invention is not limited thereto. For example, the pair of jaw members may be driven by rigid elongate elements such as rods or the like.

(130) Further, in first and second embodiments described above, the case has been described in which the operation handle including the pair of grip members is provided as the input device for inputting the command to open and close the pair of jaw members. However, the invention is not limited thereto. For example, an operation handle other than the operation handle that includes the pair of grip members may be provided as an input device for inputting the command to open and close the pair of jaw members. Also, a device or the like other than the operation handle may be provided as an input device for inputting the command to open and close the pair of jaw members.

(131) Further, in a first embodiment described above, the case has been described in which the predetermined value, which is, the difference between the value of the command jaw opening angle when the motor current value of the drive part exceeds the predetermined threshold value and the minimum value of the command jaw opening angle in the restriction mode is the same as the difference between the value of the command jaw opening angle when the opening angle between the pair of jaw members is zero and the initial minimum value of the command jaw opening angle. For example, the predetermined value may be a value larger than or smaller than the difference between the value of the jaw command opening angle when the opening angle between the pair of jaw members is zero and the initial minimum value of the jaw command opening angle.

(132) Further, in first and second embodiments described above, the case has been described in which the controller is configured, when it is determined that the current value of the drive part exceeds the predetermined threshold value and the value of the command jaw opening angle gets smaller than the previous value of the command jaw opening angle, to switch to the restriction mode. However, the invention is not limited thereto. For example, the controller may be configured, when it is detected that the motor current value of the drive part exceeds the predetermined threshold value, to switch to the restriction mode regardless of the value of the command jaw opening angle.

(133) Further, in first and second embodiments described above, the case has been described in which the controller is configured, when it is repeatedly determined multiple times that the current value of the drive part exceeds the predetermined threshold value and the value of the command jaw opening angle gets smaller than the previous value of the command jaw opening angle, to switch to the restriction mode. However, the invention is not limited thereto. For example, the controller may be configured, when it is determined only once that the current value of the drive part exceeds the predetermined threshold value and the value of the command jaw opening angle

gets smaller than the previous value of the command jaw opening angle, to switch to the restriction mode.

(134) Further, in first and second embodiments described above, the case has been described in which the controller is configured to release the restriction mode, when the motor current values of the drive parts get equal to or less than the predetermined threshold value and when the value of the command jaw opening angle generated exceeds the value of the command jaw opening angle at the start of the restriction mode or the value close to the maximum value of the command jaw opening angle (the approximate maximum value of the command jaw opening angle). However, the invention is not limited thereto. For example, the controller may be configured, when it is determined that the motor current values of the drive parts get equal to or less than the predetermined threshold value in the following state, to release the restriction mode regardless of the value of the command jaw opening angle.

(135) In first and second embodiments described above, the case has been described in which the predetermined threshold value is set according to the above described equation (1). However, the invention is not limited thereto. For example, the predetermined threshold value may be set based on an equation other than the above described equation (1).

(136) In first and second embodiments described above, the case has been described in which the fall-off prevention part is provided. However, the disclosure is not limited thereto. For example, the fall-off prevention part may not be provided.

(137) In one or more embodiments described above, the case has been described in which the robotic surgical system is configured as a master-slave system. However, the disclosure is not limited thereto. For example, the robotic surgical system may be a system including a semi-automated process in which the patient-side apparatus automatically performs a part of the process without receiving instructions from the remote control apparatus, or including a fully automated process in which the patient-side apparatus performs the process independently. In such a case, the control unit or the controller generates the command jaw opening angle based on a control program including the restriction mode.

(138) In a second embodiment described above, the case has been described in which the upper limit of the minimum value of the command jaw opening angle is set in the restriction mode. However, the invention is not limited thereto. For example, it is not necessary to set the upper limit of the minimum value of the jaw command opening angle in the restriction mode.

(139) In a second embodiment described above, the case has been described in which the upper limit of the minimum value of the command jaw opening angle is a negative value smaller than zero by the predetermined amount. However, the invention is not limited to this. For example, the upper limit may be zero.

(140) The functions of each of the elements disclosed herein may be carried out by a circuitry or a processing circuitry including a general purpose processor, a dedicated processor, an integrated circuit, an ASIC (Application Special Integrated Circuit), a conventional circuit, or a combination of two or more of them, that is configured or programmed to perform the functions. A processor is considered a processing circuitry or a circuitry because it contains transistors and other circuit elements. In the disclosure, a circuit, a unit, or a means may be either a hardware that is configured to perform the recited function(s) or a hardware that is programmed to perform the recited function(s). The hardware may be the hardware disclosed herein, or may be other known hardware that is programmed or configured to perform the function(s) described. If the hardware is a processor which is considered as a type of a circuit, a circuit, a means, or a unit is a combination of hardware and software, and the software is used to configure the hardware and/or the processor.

Claims

1. A robotic surgical system comprising: a surgical instrument including a pair of jaw members configured to be driven by elongate elements connected to a driven member so as to be opened and closed to grasp an object; an input device to which a command for opening and closing the pair of jaw members is to be input; and a surgical robot that includes a robot arm to which the surgical instrument is attached and which includes a drive part configured to drive the driven member, and a controller configured to drive the drive part based on a command jaw opening angle associated with an input to the input device for controlling an opening angle between the pair of jaw members, wherein the controller is configured to determine whether or not a current value of the drive part exceeds a predetermined threshold value during a closing operation of the pair of jaw members and, when it is determined that the current value of the drive part exceeds the predetermined threshold value, to drive the drive part in a restriction mode in which a magnitude of the command jaw opening angle is restricted.
2. The robotic surgical system according to claim 1, wherein the controller is configured, when it is determined that the current value of the drive part exceeds the predetermined threshold value during the closing operation of the pair of jaw members, to drive the drive part in the restriction mode in which a minimum value of the command jaw opening angle is restricted to a value greater than an initial minimum value of the command jaw opening angle, which is the minimum value of the command jaw opening angle in a mode other than the restriction mode.
3. The robotic surgical system according to claim 2, wherein a difference between the value of the command jaw opening angle when the current value of the drive part exceeds the predetermined threshold value and the minimum value of the command jaw opening angle in the restriction mode is set same as a difference between the value of the command jaw opening angle that makes the opening angle between the pair of jaw members zero degrees and the initial minimum value.
4. The robotic surgical system according to claim 1, further comprising a storage configured to store the value of the command jaw opening angle at predetermined intervals, wherein the controller is configured, when it is determined that the current value of the drive part exceeds the predetermined threshold value and the value of the command jaw opening angle gets smaller than the value of the command jaw opening angle that is stored in the storage, to switch to the restriction mode.
5. The robotic surgical system according to claim 4, wherein the controller is configured, when it is repeatedly determined multiple times that the current value of the drive part exceeds the predetermined threshold value and the value of the command jaw opening angle gets smaller than the value of the command jaw opening angle that is stored in the storage, to switch to the restriction mode.
6. The robotic surgical system according to claim 1, wherein the controller is configured, when the current value of the drive part gets equal to or smaller than the predetermined threshold value, to release the restriction mode.
7. The robotic surgical system according to claim 6, wherein in a state where the surgical instrument is operable by the input device, the controller is configured to release the restriction mode, when the current value of the drive part gets equal to or less than the predetermined threshold value and the value of the command jaw opening angle exceeds the value of the command jaw opening angle at a start of the restriction mode or a value close to a maximum value of the command jaw opening angle.
8. The robotic surgical system according to claim 7, wherein the surgical robot includes a second robot arm to which an endoscope is to be attached, the robotic surgical system further comprises a display configured to display an image captured by the endoscope, the display includes a sensor configured to detect a presence of an operator, the controller is configured, when the sensor of the display detects the presence of the operator, to determine that the surgical instrument is operable by the input device.

9. The robotic surgical system according to claim 6, wherein in a state where the surgical instrument is not operable by the input device, the controller is configured, when the current value of the drive part gets equal to or smaller than the predetermined threshold value, to release the restriction mode regardless of the value of the command jaw opening angle.

10. The robotic surgical system according to claim 9, wherein the surgical robot includes a second robot arm to which an endoscope is to be attached, the robotic surgical system further comprises a display configured to display an image captured by the endoscope, the display includes a sensor configured to detect a presence of an operator, the controller is configured, when the sensor of the display does not detect the presence of the operator, to determine that the surgical instrument is not operable by the input device.

11. The robotic surgical system according to claim 1, wherein the predetermined threshold value is set based on a frictional force of the drive part.

12. The robotic surgical system according to claim 1, wherein the elongate elements are composed of wires or cables, the surgical instrument further includes pulleys around which the elongate elements to drive the pair of jaw members are wound, the pulleys include engagement portions engaged with the pair of jaw members for driving the jaw members, and the pair of jaw members is configured, when the pulleys are rotated about a rotational axis of the pulleys, to be opened and closed about a rotational axis different from the rotational axis of the pulleys.

13. The robotic surgical system according to claim 2, wherein the controller is configured, when it is determined that the current value of the drive part exceeds the predetermined threshold value during the closing operation of the pair of jaw members, to drive the drive part in the restriction mode in which the minimum value of the command jaw opening angle is restricted to the value greater than the initial minimum value of the command jaw opening angle and not greater than zero.

14. The robotic surgical system according to claim 13, wherein an upper limit of the minimum value of the commanded jaw opening angle in the restriction mode is set to a negative value less than zero by a predetermined amount.

15. The surgical robot according to claim 13, wherein the controller is configured to detect the opening angle between the pair of jaw members at a time when the current value of the drive part exceeds the predetermined threshold value, and configured not to switch to the restriction mode, when the detected opening angle between the pair of jaw members is less than a predetermined value even when it is determined that the current value of the drive part exceeds the predetermined threshold value during the closing operation of the pair of jaw members.

16. A robotic surgical method for a robotic surgical system, the robotic surgical system comprising: a surgical instrument including a pair of jaw members configured to be driven by elongate elements connected to a driven member so as to be opened and closed to grasp an object; an input device to which a command for opening and closing the pair of jaw members is to be input; and a surgical robot including a robot arm to which the surgical instrument is attached and which includes a drive part configured to drive the driven member, the robotic surgical method comprising: obtaining a command jaw opening angle that is associated with an input to the input device for controlling an opening angle between the pair of jaw members; and driving the drive part based on the command jaw opening angle, wherein driving the drive part comprises: determining whether or not a current value of the drive part exceeds a predetermined threshold value during a closing operation of the pair of jaw members; and when it is determined that the current value of the drive part exceeds the predetermined threshold value, driving the drive part in a restriction mode in which a magnitude of the command jaw opening angle is restricted.

17. A surgical robot comprising: a surgical instrument including a pair of jaw members configured to be driven by elongate elements connected to a driven member so as to be opened and closed to grasp an object; a robot arm to which the surgical instrument is attached and including a drive part configured to drive the driven member; a controller configured to drive the drive part based on a

command jaw opening angle for an opening angle between the pair of jaw members, wherein the controller is configured to determine whether or not a current value of the drive part exceeds a predetermined threshold value during a closing operation of the pair of jaw members, and when it is determined that the current value of the drive part exceeds the predetermined threshold value, to drive the drive part in a restriction mode in which a magnitude of the command jaw opening angle is restricted.

18. The surgical robot according to claim 17, wherein the controller is configured, when it is determined that the current value of the drive part exceeds the predetermined threshold value, to drive the drive part in the restriction mode in which a minimum value of the command jaw opening angle is restricted to a value greater than an initial minimum value of the command jaw opening angle, which is the minimum value of the command jaw opening angle in a mode other than the restriction mode.

19. The surgical robot according to claim 18, wherein a difference between the value of the command jaw opening angle when the current value of the drive part exceeds the predetermined threshold value and the minimum value of the command jaw opening angle in the restriction mode is set same as a difference between the value of the command jaw opening angle that makes the opening angle between the pair of jaw members zero degrees and the initial minimum value.

20. The surgical robot according to claim 17, further comprising a storage configured to store the value of the command jaw opening angle at predetermined intervals, wherein the controller is configured, when it is determined that the current value of the drive part exceeds the predetermined threshold value and the value of the command jaw opening angle gets smaller than the value of the command jaw opening angle that is stored in the storage, to switch to the restriction mode.
